

School Management System

EduSchool - The Next-Gen High School Management System

- **Student Name:** MENH Chanheng (2023445), HENG Shito (2023210), PIN Sovanda (2023457)
- **Supervisor:** CHANDAN MUKHERJEE

Table of Contents

1	Introduction	2
2	Objective	2
3	Project Planning and Scheduling	3
4	Development methodology	3
5	Technologies /Tools	3
6	System Design	4
7	Database Design	5
8	Modularization details (Project Functionality)	6
9	Validation Check	13
10	Special Features	14
11	Testing and Results	16
12	Future Work	23
13	References	23
14	Individual Contribution	24

Introduction

EduSchool is a centralized web-based platform designed to modernize high school administration.

- **Problem Statement:** Many high schools rely on fragmented manual systems, leading to data redundancy and communication gaps between teachers and parents.
- **Motivation:** Automating administrative tasks allows educators to focus on teaching rather than paperwork while providing parents with real-time insights into student progress.

Objective

Functional Objectives:

- Develop a centralized database for student, teacher, and staff records.
- Automate grade calculation and report card generation.
- Provide secure login portals for Admin, Teacher, and Student/Parent roles.

Non-Functional Objectives:

- **Security:** Implement password encryption and Role-Based Access Control (RBAC).
- **Scalability:** Ensure the system can handle records for up to 2,000 students.

Project Planning and Scheduling

Phase	Tasks	Duration	Deadline
Research & Planning	Literature review, requirement analysis	2 weeks	26/01/2026
Design	System architecture, UI/UX design	2 weeks	09/02/2026
Implementation	Coding, Integration, Testing	6 weeks	23/03/2026
Testing & Debugging	Unit testing, System testing	2 weeks	06/04/2026
Documentation & Final Report	Writing project documentation	2 weeks	20/04/2026

Development methodology

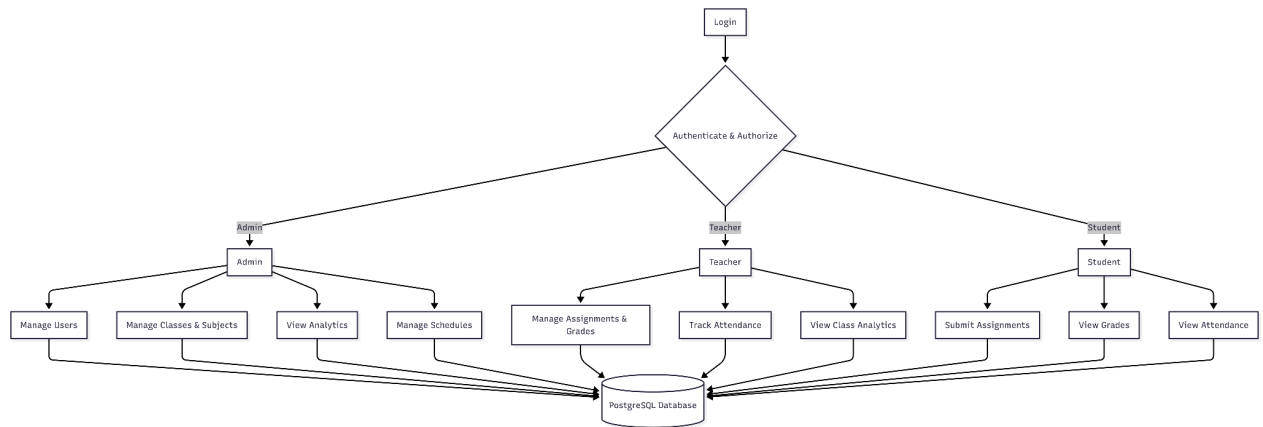
The project utilizes the **Agile** framework, working in one-month sprints to allow for iterative testing and continuous feedback

Technologies/Tools

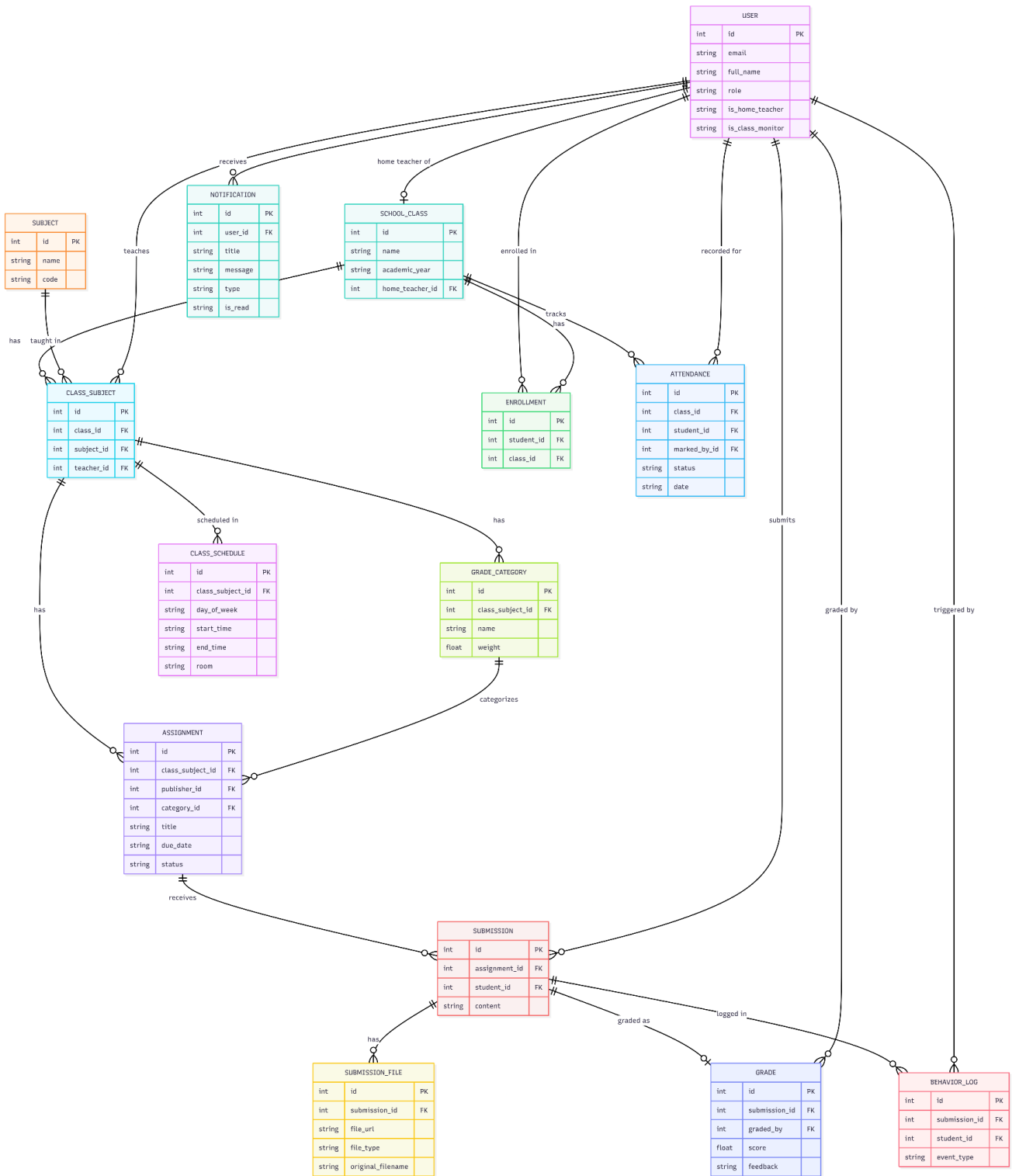
- **Frontend:** Next.js 14 (App Router).
- **Backend Logic:** FastAPI (Python).
- **Database:** PostgreSQL hosted on Supabase.
- **Deployment:** Vercel for the frontend and Railway/Render for the API.

System Design

- Authentication Design

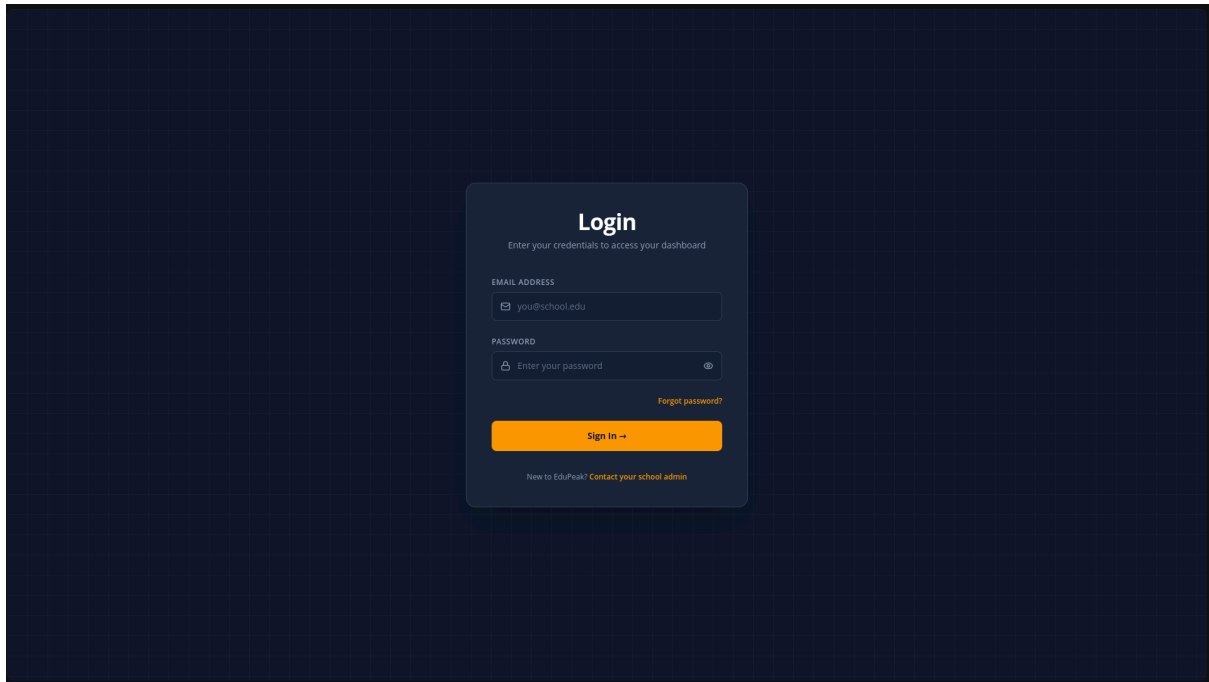


Database design

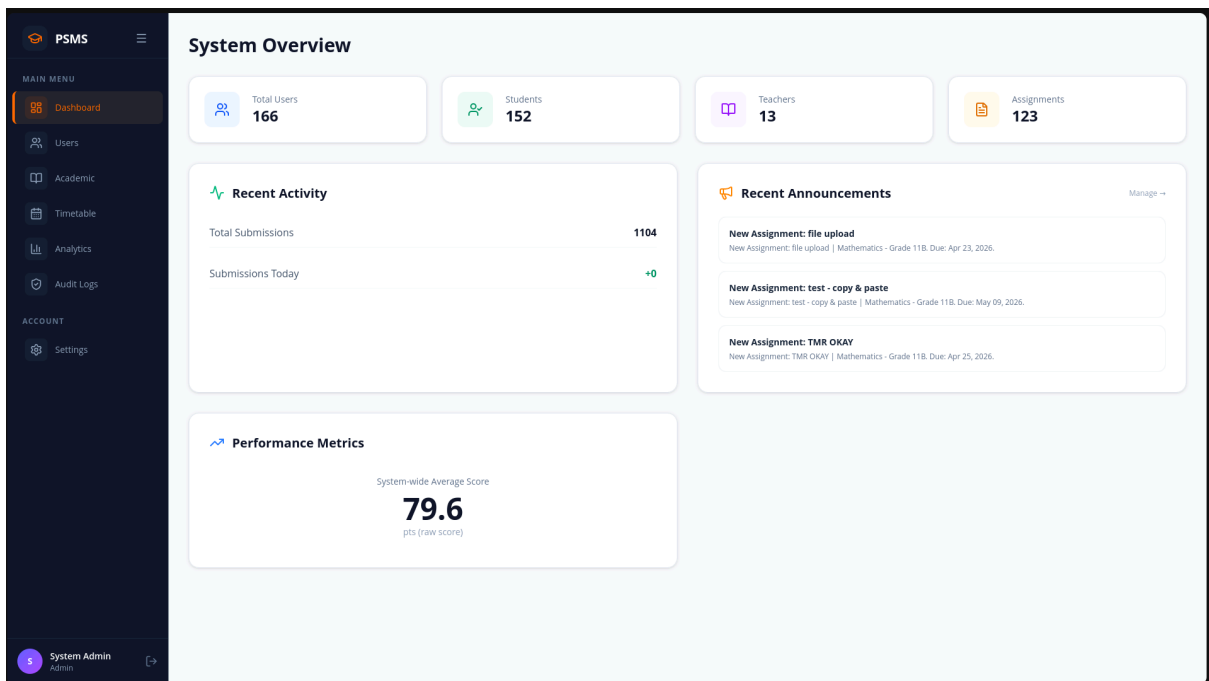


Modularization details (Project Functionality)

- Login Page



- Admin Page



PSMS

MAIN MENU

Dashboard

Users

Academic

Timetable

Analytics

Audit Logs

ACCOUNT

Settings

System Admin

Admin

User Management

Manage all system accounts and roles.

Import CSV

Add User

TOTAL USERS

169

STUDENTS

155

TEACHERS

13

ADMINS

1

Search by name or email...

All Roles

#	Name	Email	Role	Gender	Status	Flags	Created	Actions
1	System Admin	admin@srmk.edu.kh	admin	—	Active	—	Apr 20, 2026	EditReset PWDelte
2	Socheat Sok	socheat.sok@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
3	Dara Heng	dara.heng@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
4	Vibol Tep	vibol.tep@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
5	Leakena Ros	leakena.ros@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
6	Chantha Iim	chantha.iim@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
7	Makara Chea	makara.chea@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
8	Chenda Sen	chenda.sen@srmk.edu.kh	teacher	—	Active	—	Apr 20, 2026	EditReset PWDelte
9	Sopheha Lim	2025001lim@srmk.edu.kh	student	—	Active	CM	Apr 20, 2026	EditReset PWDelte
10	Sreymom Heng	2025003heng@srmk.edu.kh	student	—	Active	—	Apr 20, 2026	EditReset PWDelte
11	Vanina Nget	2025004nget@srmk.edu.kh	student	—	Active	—	Apr 20, 2026	EditReset PWDelte
12	Ratana Kong	2025005kong@srmk.edu.kh	student	—	Active	—	Apr 20, 2026	EditReset PWDelte
13	Bopha Chea	2025006chea@srmk.edu.kh	student	—	Active	—	Apr 20, 2026	EditReset PWDelte
14	Sokha Tep	2025007tep@srmk.edu.kh	student	—	Active	—	Apr 20, 2026	EditReset PWDelte

PSMS

MAIN MENU

Dashboard

Users

Academic

Timetable

Analytics

Audit Logs

ACCOUNT

Settings

System Admin

Admin

Academic Management

Manage classes and subjects offered in the system.

Bulk Create

Add Class

Classes

Subjects

Subject Sets

Cohort Setup

5 classes total

ID	Name	Academic Year	Home Teacher	Actions
1	Grade 11A	2025-2026	Sovann Keo	Delte
2	Grade 11B	2025-2026	Sreymom Mao	Delte
3	Grade 11C	2025-2026	Piseth Rath	Delte
4	Grade 11D	2025-2026	Chanthy Noun	Delte
5	Grade 11E	2025-2026	Kunthea Lim	Delte

PSMS

MAIN MENU

Dashboard

Users

Academic

Timetable

Analytics

Audit Logs

ACCOUNT

Settings

System Admin
Admin

Timetable

Master schedules for all Grade 11 classes

View by Class

Grade 11A

2025-2026

+ Add Draft Block

Publish (0)

	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
7:00 AM	Khmer 2h LIVE	Biology 1h 30m LIVE	Mathematics 2h LIVE	Economy 1h 30m LIVE	Physics 1h 30m LIVE	Geography 1h 30m LIVE
8:00 AM	Chanthy Noun Grade 11A	Socheat Sok Grade 11A	Sovann Kao Grade 11A	Chantha Im Grade 11A	Piseth Rath Grade 11A	Vibol Tep Grade 11A
9:00 AM	Mathematics 2h LIVE	English 1h 30m LIVE	Khmer 2h LIVE	Earth Science 1h 30m LIVE	Chemistry 1h 30m LIVE	Morality & Civ... 1h 30m LIVE
10:00 AM	Sovann Kao Grade 11A	Sreymom Mao Grade 11A	Chanthy Noun Grade 11A	Chenda Sen Grade 11A	Bopha Pichss Grade 11A	Dara Heng Grade 11A
11:00 AM						Art 1h 30m LIVE
12:00 PM						Leakena Ros Grade 11A
1:00 PM	Physics 1h 30m LIVE	History 1h 30m LIVE	Geography 1h 30m LIVE	Physical Educ... 1h 30m LIVE	English 1h 30m LIVE	Economy 1h 30m LIVE
2:00 PM	Piseth Rath Grade 11A	Kunthea Lim Grade 11A	Vibol Tep Grade 11A	Makara Chea Grade 11A	Sreymom Mao Grade 11A	Chantha Im Grade 11A
3:00 PM	Chemistry 1h 30m LIVE	Art 1h 30m LIVE	Morality & Civ... 1h 30m LIVE	Biology 1h 30m LIVE	History 1h 30m LIVE	Earth Science 1h 30m LIVE
	Bopha Pichss	Leakena Ros	Dara Heng	Socheat Sok	Kunthea Lim	Chenda Sen

PSMS

MAIN MENU

Dashboard

Users

Academic

Timetable

Analytics

Audit Logs

ACCOUNT

Settings

System Admin
Admin

System Analytics

Platform-wide overview and per-class performance data

System Overview

TOTAL USERS

169

STUDENTS

155

TEACHERS

13

ASSIGNMENTS

124

SUBMISSIONS

1106

SUBMISSIONS TODAY

2

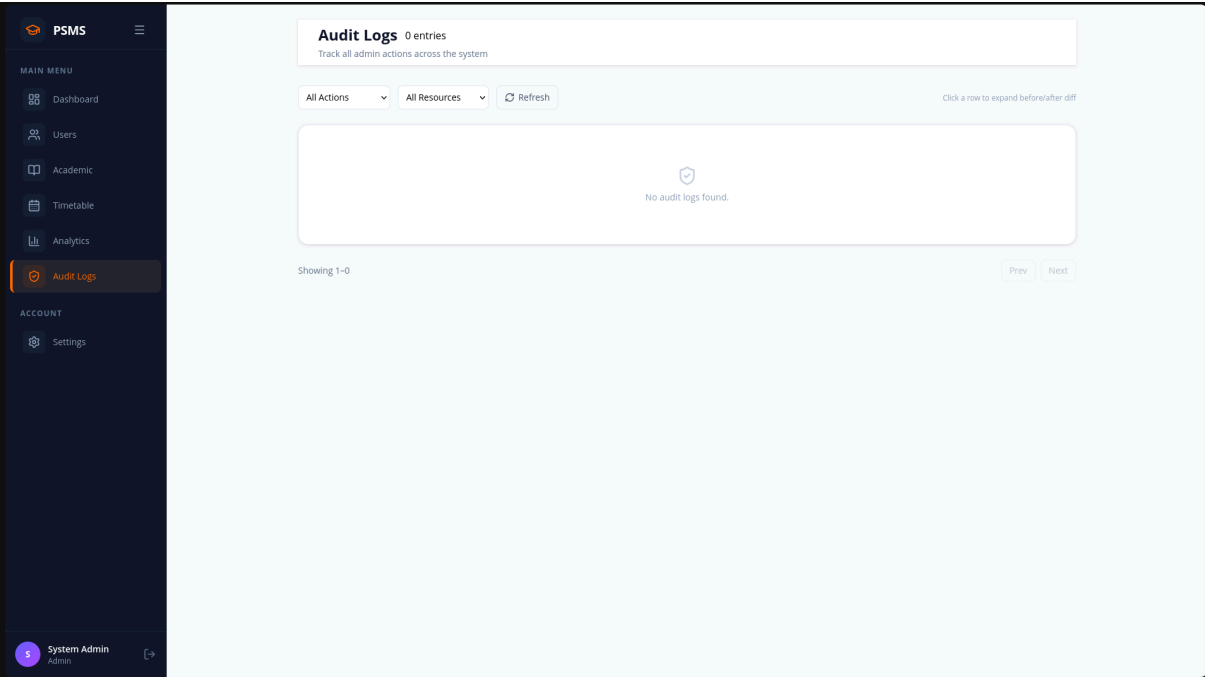
SYSTEM AVG SCORE

79.5

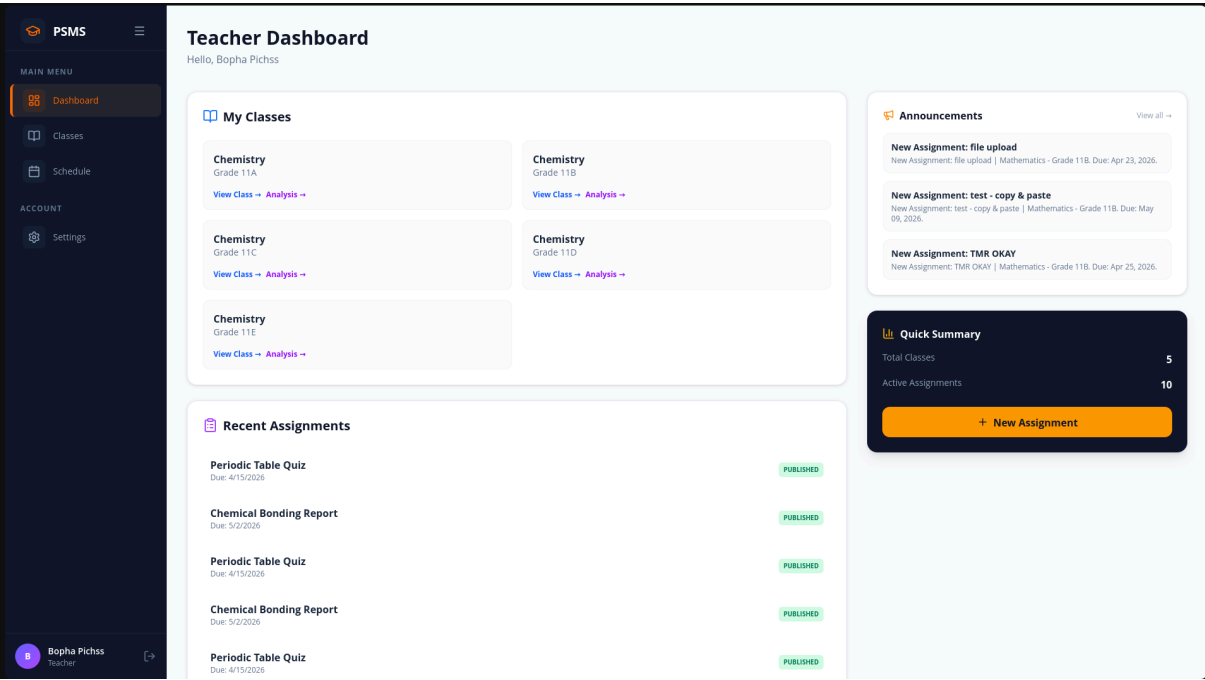
Class Analytics

Select a class: -- Choose a class --

Select a class above to view its analytics.



● **Teacher Page**



PSMS

MAIN MENU

Dashboard

Classes

Schedule

ACCOUNT

Settings

Bopha Pichss

Teacher

My Classes

Manage your assigned class subjects

5 classes assigned

Chemistry

Grade 11A

Teacher

Chemistry

Grade 11B

Teacher

Chemistry

Grade 11C

Teacher

Chemistry

Grade 11D

Teacher

Chemistry

Grade 11E

Teacher

PSMS

MAIN MENU

Dashboard

Classes

Schedule

ACCOUNT

Settings

Bopha Pichss

Teacher

My Schedule

Your teaching schedule for the current term

10 sessions per week

MONDAY

Chemistry

Grade 11B

2:30 PM - 4:00 PM

Grade 11B

Chemistry

Grade 11D

2:30 PM - 4:00 PM

Grade 11D

Chemistry

Grade 11C

2:30 PM - 4:00 PM

Grade 11C

Chemistry

Grade 11E

2:30 PM - 4:00 PM

Grade 11E

Chemistry

Grade 11A

2:30 PM - 4:00 PM

Grade 11A

FRIDAY

Chemistry

Grade 11E

8:30 AM - 10:00 AM

Grade 11E

Chemistry

Grade 11A

8:30 AM - 10:00 AM

Grade 11A

Chemistry

Grade 11B

8:30 AM - 10:00 AM

Grade 11B

Chemistry

Grade 11C

8:30 AM - 10:00 AM

Grade 11C

Chemistry

Grade 11D

8:30 AM - 10:00 AM

Grade 11D

- Student Page

PSMS

MAIN MENU

Dashboard

Classes

Schedule

Notifications 10

ACCOUNT

Settings

Piseth Mao
Student

Welcome, Piseth Mao

Saturday, April 25, 2026

70.92
Avg Score
This Term

1
Active Classes
Enrolled

N/A
Attendance
Tracked Daily

22
Published Tasks
In all subjects

My Subjects

Biology
Socheat Sok

Morality & Civilization
Dara Heng

Geography
Vibol Tep

Art
Leakana Ros

Economy
Chantha Im

Physical Education
Makara Chea

Earth Science
Chenda Sen

Chemistry
Bopha Pichs

Mathematics
Sovann Keo

Physics
Piseth Rath

English
Sreymom Mao

Khmer
Chanthy Noun

History
Kunthea Lim

Active Assignments

☐ Linear Equations — Chapter 1 Problem Set
Max Score: 100
Due: 4/6/2026

☐ Quadratic Functions Quiz
Max Score: 50
Due: 4/30/2026

PSMS

MAIN MENU

Dashboard

Classes

Schedule

Notifications 10

ACCOUNT

Settings

Piseth Mao
Student

My Classes

13 Active Courses

Biology

Socheat Sok

Active

View Course →

Morality & Civilization

Dara Heng

Active

View Course →

Geography

Vibol Tep

Active

View Course →

Art

Leakana Ros

Active

View Course →

Economy

Chantha Im

Active

View Course →

Physical Education

Makara Chea

Active

View Course →

Earth Science

Chenda Sen

Active

View Course →

Chemistry

Bopha Pichs

Active

View Course →

Mathematics

Sovann Keo

Active

View Course →

Physics

Piseth Rath

Active

View Course →

English

Sreymom Mao

Active

View Course →

Khmer

Chanthy Noun

Active

View Course →

PSMS

MAIN MENU

Dashboard

Classes

Schedule

Notifications

ACCOUNT

Settings

Pieth Mao Student

Schedule

Weekly Timetable

26 classes scheduled

Week

Day

Time / Day	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
7:00 AM	Khmer Chanthy Noun 7:00 AM - 9:00 AM @ Grade 11A	Biology Socheat Sok 7:00 AM - 8:30 AM @ Grade 11A	Mathematics Sovann Keo 7:00 AM - 9:00 AM @ Grade 11A	Economy Chantha Im 7:00 AM - 8:30 AM @ Grade 11A	Physics Pieth Rath 7:00 AM - 8:30 AM @ Grade 11A
8:00 AM		English Sreymom Mao 8:30 AM - 10:00 AM @ Grade 11A		Earth Science Chenda Sen 8:30 AM - 10:00 AM @ Grade 11A	Chemistry Bopha Pichs 8:30 AM - 10:00 AM @ Grade 11A
9:00 AM	Mathematics Sovann Keo 9:00 AM - 11:00 AM @ Grade 11A		Khmer Chanthy Noun 9:00 AM - 11:00 AM @ Grade 11A		
10:00 AM					
1:00 PM	Physics Pieth Rath 1:00 PM - 2:30 PM @ Grade 11A	History Kunthea Lim 1:00 PM - 2:30 PM @ Grade 11A	Geography Vibol Tip 1:00 PM - 2:30 PM @ Grade 11A	Physical Education Makara Chea 1:00 PM - 2:30 PM @ Grade 11A	English Sreymom Mao 1:00 PM - 2:30 PM @ Grade 11A
2:00 PM	Chemistry Bopha Pichs 2:30 PM - 4:00 PM @ Grade 11A	Art Leakiara Ros 2:30 PM - 4:00 PM @ Grade 11A	Morality & Civilization Dara Heng 2:30 PM - 4:00 PM @ Grade 11A	Biology Socheat Sok 2:30 PM - 4:00 PM @ Grade 11A	History Kunthea Lim 2:30 PM - 4:00 PM @ Grade 11A
4:00 PM					

PSMS

MAIN MENU

Dashboard

Classes

Schedule

Notifications

ACCOUNT

Settings

P

Piseth Mao

Student

Notifications

7 unread messages

Mark all as read

New Assignment: Final-101

New Assignment: Final-101 | Mathematics - Grade 11A. Due: Apr 21, 2026.

From: Sovann Keo

5 days ago

Assignment Due Tomorrow

Reminder: 'Chapter 3 Exercises' in Mathematics is due tomorrow at 11:59 PM.

From: Sovann Keo

5 days ago

Grade Posted: Quiz 1

Your grade for 'Chapter 1 Quiz' in Mathematics has been posted.

From: System

5 days ago

Grade Posted: Essay Submission

Your English essay has been graded. Check the Grades tab for your score and feedback.

From: Sreymom Mao

5 days ago

New Assignment: Lab Report

A new assignment 'Refraction Lab Report' has been posted in Physics. Due in 3 days.

From: Piseth Rath

5 days ago

Library Books Due Reminder

Please return any overdue library books by Friday to avoid late fees.

From: Librarian

5 days ago

Welcome to the New Term

Welcome back! The new academic term has started. Check your enrolled classes and upcoming assignments.

From: System

5 days ago

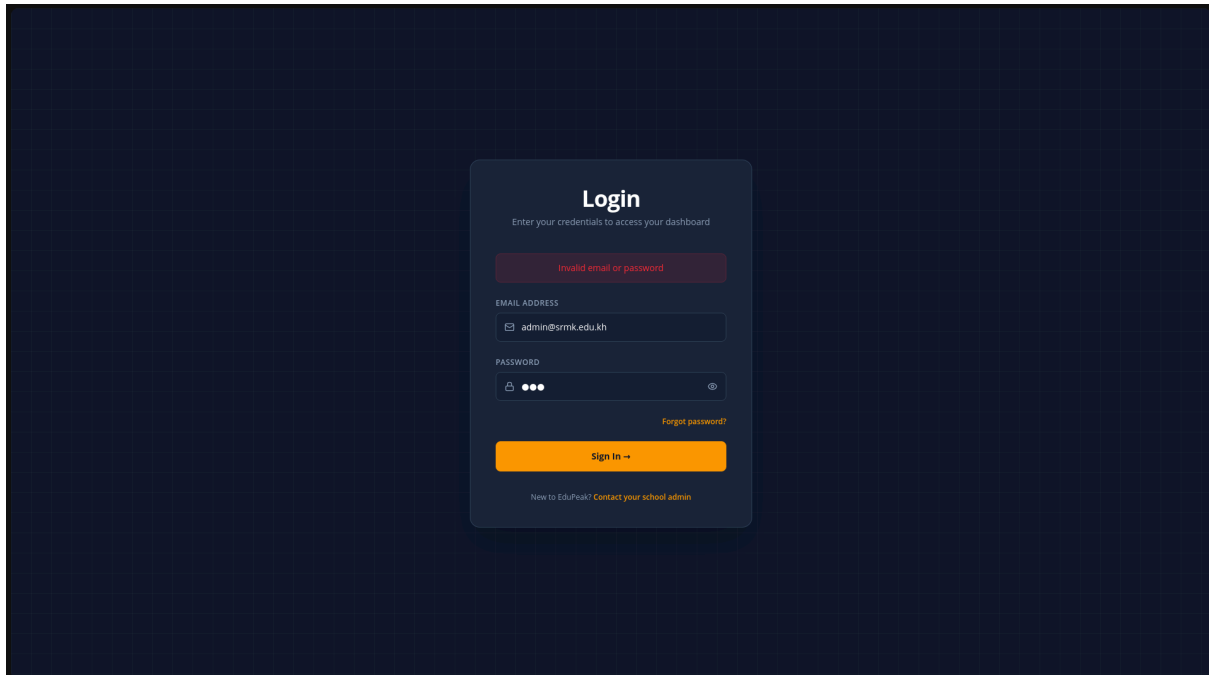
Mid-Term Exam Schedule Updated

The mid-term exam timetable has been revised. Please review the new schedule in your class portal.

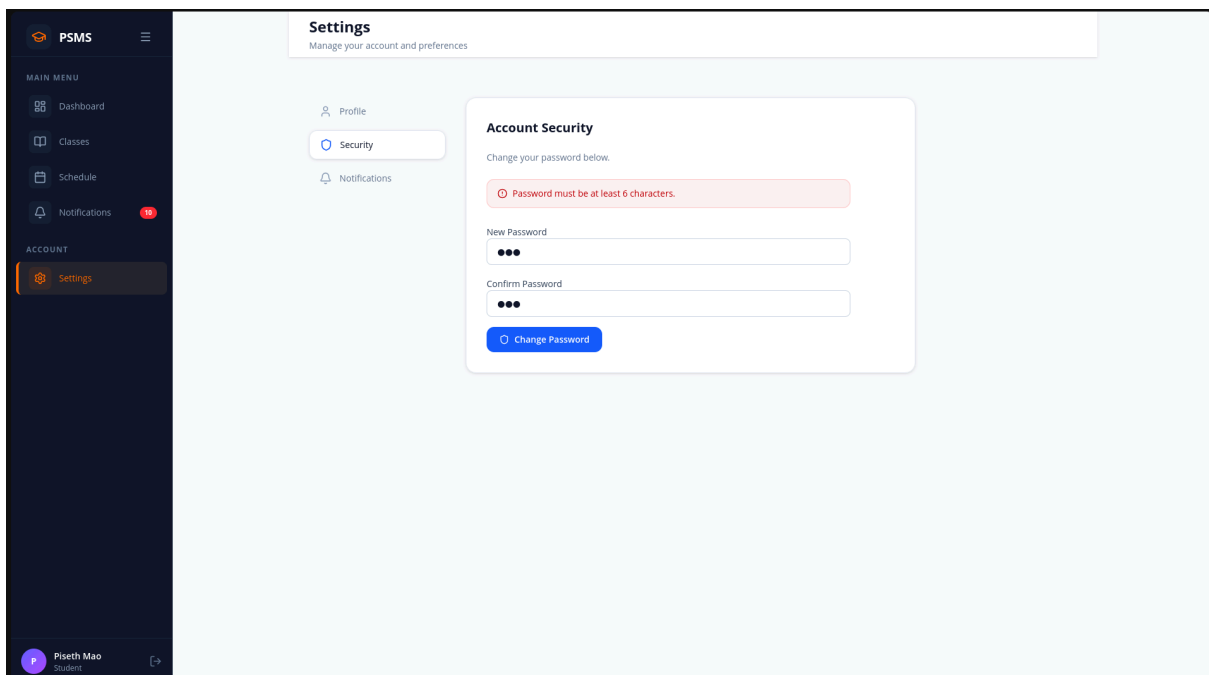
5 days ago

Validation Check

- Check if email or password valid

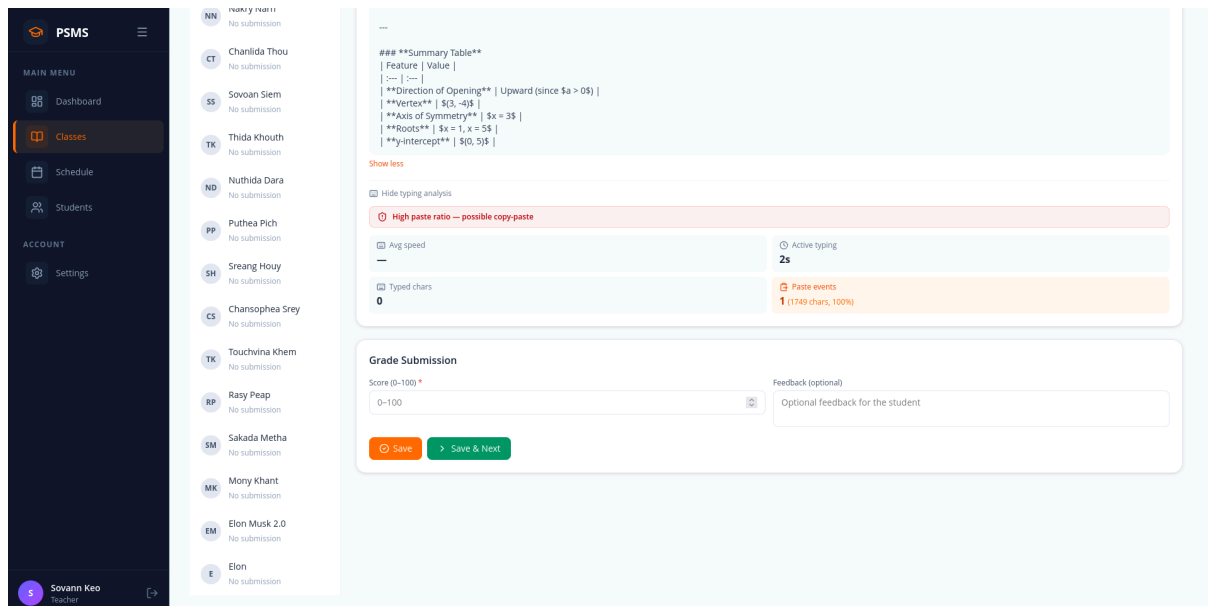


- Password Reset Checking: Minimum of 6 characters

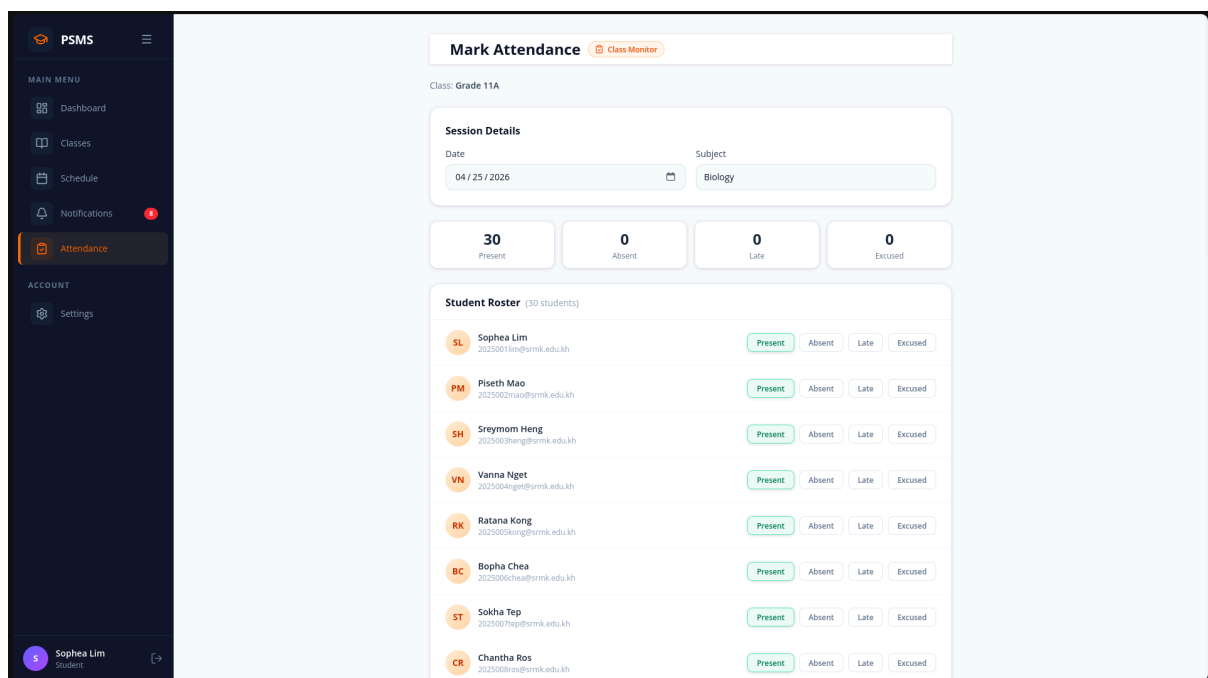


Special Features

1. **Copy-Paste Detection:** Teachers could track how students work with their assignment by their own typing, or copy-paste from other sources (friends, AI, ...)



2. **Student Attendance Drafting:** It's the special feature only available for the student assigned to be the class monitor as he/she could draft the attendance (present/absent/late/excused) for other students beside teachers.



3. **Analytics Feature:** It's the special feature for teacher, whose role is the home teacher for each class to observe the performance of each students in each subjects of his/her classroom

PSMS

Grade 11A Home Teacher
Academic Year: 2025-2026

TOTAL STUDENTS 33

RANKED STUDENTS 33 / 33

Student Rankings

RANK	NAME	EMAIL	ROLE	TOTAL SCORE	ACTIVITY
1	S Sreynich Khen	2025018khen@srmk.edu.kh	Student	913.0 avg 83.0	Logs
2	K Kunthea Im	2025009im@srmk.edu.kh	Student	910.0 avg 82.7	Logs
3	C Chenda Bun	2025014bun@srmk.edu.kh	Student	905.0 avg 82.3	Logs
4	R Ratana Kong	2025005kong@srmk.edu.kh	Student	902.0 avg 82.0	Logs
5	M Makara Sen	2025010sen@srmk.edu.kh	Student	897.0 avg 81.5	Logs
6	S Sophea Lim	2025001lim@srmk.edu.kh	Class Monitor	894.0 avg 81.3	Logs
7	C Chantha Ros	2025008ros@srmk.edu.kh	Student	893.0 avg 81.2	Logs
8	S Samoan Nien	2025013nien@srmk.edu.kh	Student	888.0 avg 80.7	Logs
9	V Vanna Nget	2025004nget@srmk.edu.kh	Student	885.0 avg 80.5	Logs

Sovann Keo
Teacher

Testing and Results

- Testing methodologies (e.g., unit testing, integration testing).

1. Unit Testing: Boundary Grade Test:

Endpoint Tested: PUT /grades/{grade_id}

Findings: The system was tested using "Out of Bounds" values to determine if the backend enforces logical academic constraints. Two specific scenarios were executed:

- ❖ **Upper Boundary Test:** An input of **105** was accepted and processed as a valid score.
- ❖ **Lower Boundary Test:** An input of **-5** was accepted and processed as a valid score

Status: ❌ **FAIL (Constraint Missing)**

PUT /grades/{grade_id} Update Grade

Parameters

Name	Description
grade_id * required integer (path)	1

Request body required

Edit Value | Schema

```
{  "score": 105,  "feedback": "string"}
```

Code	Details
200	Response body<pre>{ "id": 1, "submission_id": 1, "score": 105, "feedback": "string", "graded_by": 2, "graded_at": "2026-04-20T13:57:43.907136+07:00"}</pre> Response headers<pre>access-control-allow-credentials: true content-length: 121 content-type: application/json date: Tue, 21 Apr 2026 17:16:41 GMT server: uvicorn</pre>

PUT /grades/{grade_id} Update Grade

Parameters

Name	Description
grade_id * required integer (path)	1

Request body required

Edit Value | Schema

```
{  "score": -5,  "feedback": "string"}
```

Server response	
Code	Details
200	Response body<pre>{ "id": 1, "submission_id": 1, "score": -5, "feedback": "string", "graded_by": 2, "graded_at": "2026-04-20T13:57:43.907136+07:00"}</pre> Response headers<pre>access-control-allow-credentials: true content-length: 120 content-type: application/json date: Tue, 21 Apr 2026 17:17:37 GMT server: uvicorn</pre>

2. Transaction Handshake – Data Integrity

Audit Item: Transaction Handshake & Data Integrity (Referential Integrity)

Endpoint Tested: POST /enrollments/

Findings: An attempt was made to create an enrollment record using non-existent identifiers (student_id: 9999, class_id: 8888). The system successfully blocked the transaction, returning a **500 Internal Server Error**. This confirms that PostgreSQL Foreign Key (FK) Constraints are active, ensuring that no "Orphaned Records" can exist in the system. The database acted as a final safety net to maintain 100% referential integrity.

Status:  **PASS**

POST /enrollments/ Create Enrollment

Parameters

No parameters

Request body required

Edit Value | Schema

```
{  
  "student_id": 9999,  
  "class_id": 8888  
}
```

Server response

Code	Details
500 Undocumented	Error: Internal Server Error Response body Internal Server Error Response headers content-length: 21 content-type: text/plain; charset=utf-8 date: Wed, 22 Apr 2026 14:49:01 GMT server: uvicorn

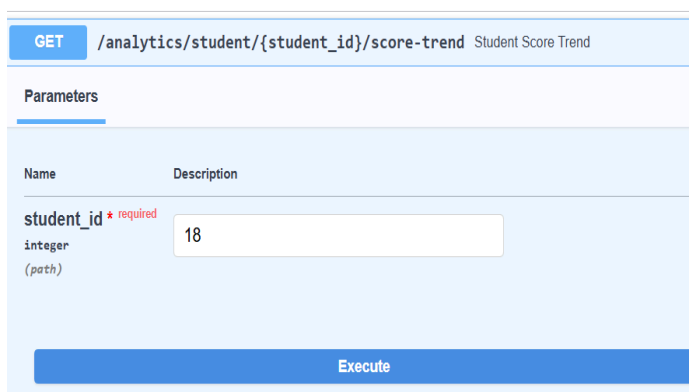
3. Test if a student can spy (view) other other student score

Audit Item: Cross-User Privacy & Data Isolation (IDOR Protection)

Endpoint Tested: /analytics/student/{student_id}/score-trend

Findings: While authenticated as a student, a request was made to view the score trends of another student (ID: 18). The system successfully intercepted the request, returning a 403 Forbidden status and the specific error message: *"Students can only view their own data."* This confirms that the identity-matching logic between the JWT token and the requested resource is fully functional.

Status:  **PASS**



Name	Description
student_id * required	
integer	18
(path)	

Execute

Server response	
Code	Details
403 <i>Undocumented</i>	Error: Forbidden
Response body	
<pre>{ "detail": "Students can only view their own data"}</pre>	
Response headers	
<pre>content-length: 50 content-type: application/json date: Wed, 22 Apr 2026 15:06:13 GMT server: uvicorn</pre>	

4. Integration Testing

Audit Item: Supabase Auth & Railways Integration

Endpoint Tested: /student -> /teacher -> /admin

Findings: Verified that a user with an assigned role is physically blocked from accessing other routes at the server level. (e.g user with admin roles, could not access to /teacher or /student routes by changing the role in url)

Status:  **PASS**

5. Role-Based Access Control (RBAC) Testing

Audit Item: Permission Boundary Verification

Findings:

User	Grade Access	Class Management	System Admin	Result
Student	Read (Own only)	No Access	No Access	✓ PASS
Teacher	Read/Edit (Assign)	Partial CRUD	No Access	✓ PASS
Admin	Full CRUD	Full CRUD	Full CRUD	✓ PASS

Status: ✓ PASS

- Performance evaluation (e.g., speed, accuracy, scalability)

1. API Response Time

Audit Item: Point-Lookup API Latency (Single Record Retrieval)

Endpoints Tested: `/users/2` and `/users/3`

Findings: Verified near-instant retrieval times of **11ms** and **14ms**. These results are significantly below the **200ms** performance threshold, confirming that the database is properly indexed and the backend-to-database "handshake" is highly optimized.

Status: ✓ PASS

All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other					
500 ms		1,000 ms		1,500 ms	
2,000 ms		2,500 ms		3,000 ms	
3,500 ms					
Name		Status	Type	Initiator	Time
2		200	fetch	swagger-ui-bundle.js:2	14 ms
3		200	fetch	swagger-ui-bundle.js:2	11 ms

2. Data accuracy: Average Calculate GPA Test:

Audit Item: GPA Calculation Accuracy (Logical Verification)

Endpoint Tested: `/analytics/student/{id}/score-trend`

Findings: Verified the mathematical aggregation of three controlled assignment scores (80, 90, and 100). The system correctly processed the total sum (270) across 3 assignments, returning a precise `average_score` and `weighted_average` of **90.0**. This confirms the backend aggregation logic is 100% accurate.

Status:  **PASS**

Response body

```
{
  {
    "assignment_id": 122,
    "assignment_title": "Test1",
    "score": 80,
    "max_score": 100,
    "submitted_at": "2026-04-22T10:57:15.852605+07:00"
  },
  {
    "assignment_id": 123,
    "assignment_title": "Test2",
    "score": 90,
    "max_score": 100,
    "submitted_at": "2026-04-22T10:57:26.069296+07:00"
  },
  {
    "assignment_id": 124,
    "assignment_title": "Test3",
    "score": 100,
    "max_score": 100,
    "submitted_at": "2026-04-22T10:57:37.007099+07:00"
  }
],
"average_score": 90,
"total_assignments": 3,
"weighted_average": 90,
"attendance_rate": 0,
"submission_rate": 11.54
}
```

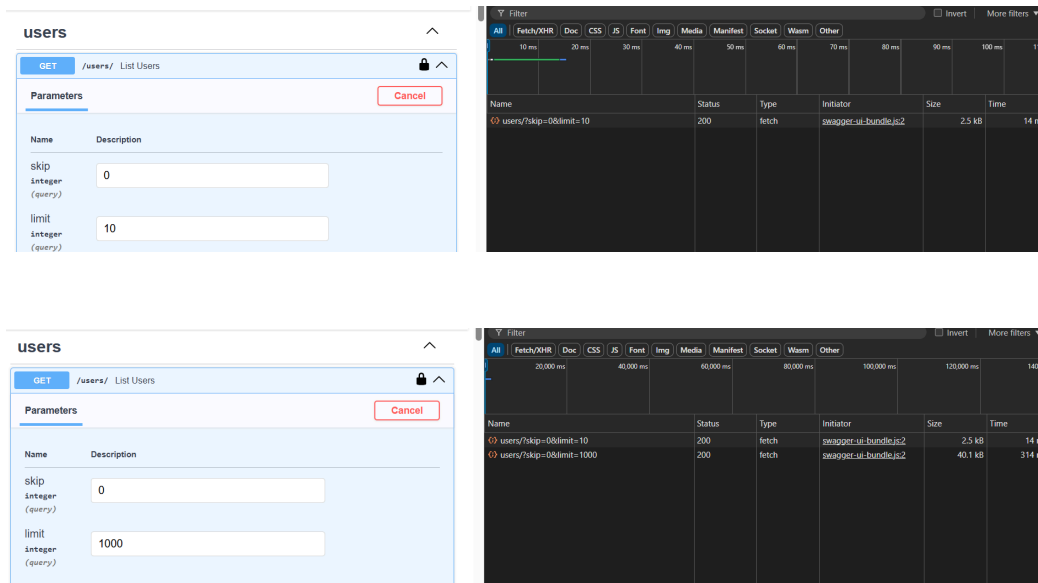
3. Scalability

Audit Item: System Scalability (High-Volume Data Retrieval)

Endpoints Tested: GET /users?limit=10 and GET /users?limit=1000

Findings: Verified the system's stability under varying data loads. A baseline request for 10 records returned in 14ms, while a high-volume stress test of 1,000 records returned in 314ms. The results confirm that an increase in data payload by 100 times results in linear, predictable latency without triggering timeouts or system instability, keeping "heavy load" responses well under the 500ms efficiency threshold.

Status:  **PASS**



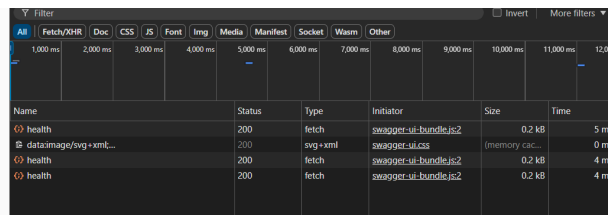
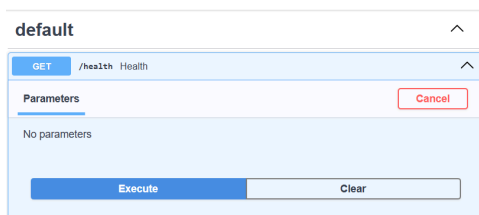
4. Cold Start (Serverless)

Audit Item: Cold Start & Serverless Responsiveness

Endpoint Tested: /health

Findings: The system was tested to measure the latency of the serverless execution environment. The initial "Cold Start" (server wake-up and database handshake) was recorded at **314ms**, while subsequent "Warm" requests returned in a record-breaking **4ms - 5ms**. Both results are significantly better than the **2.0s** industry-standard threshold for serverless applications, ensuring a seamless user experience even during the first session of the day.

Status:  **PASS**



Name	Status	Type	Initiator	Size	Time
health	200	fetch	swagger-ui_bundle.js2	0.2 kB	5 ms
data:image/svg+xml...	200	svg+xml	swagger-ui.css	(memory cac...	0 ms
health	200	fetch	swagger-ui_bundle.js2	0.2 kB	4 ms
health	200	fetch	swagger-ui_bundle.js2	0.2 kB	4 ms

Future Work

- **Payment Integration:** Adding an online fee payment gateway.
- **Validation Checking:** Improving the validate of username (alphabet only without number or symbol), email (use valid email instead of test@test), and password (checking if the password has a minimum length of 8 upwards or not) when created by admin.
- **Real-time Tracking:** Implementing school-bus GPS tracking.

References

- [1] N. C. Nweke and O. C. Ezenwa, "The Design and Implementation of Student Academic Record Management System," *ResearchGate*, 2014. [Online]. Available: https://www.researchgate.net/publication/266243197_The_Design_and_Implementation_of_Student_Academic_Record_Management_System. [Accessed: Mar. 5, 2026].
- [2] O. A. Adaramola et al., "Development of an Improved School Information Management System," *ResearchGate*, Jun. 2022. [Online]. Available: https://www.researchgate.net/publication/361017513_DEVELOPMENT_OF_AN_IMPROVED_SCHOOL_INFORMATION_MANAGEMENT_SYSTEM. [Accessed: Mar. 5, 2026].
- [3] J. Dela Cruz et al., "Intellischool: A Student Information System for Senior High School," *ResearchGate*, 2024. [Online]. Available: https://www.researchgate.net/publication/379716395_Intellischool_A_Student_Information_System_for_Senior_High_School. [Accessed: Mar. 5, 2026].
- [4] K. Beck et al., "Manifesto for Agile Software Development," *Agile Alliance*, 2001. [Online]. Available: <http://agilemanifesto.org/>. [Accessed: Mar. 5, 2026].
- [5] Vercel, "Next.js Documentation," *Next.js*. [Online]. Available: <https://nextjs.org/docs>. [Accessed: Mar. 5, 2026].

[6] S. Ramírez, "FastAPI," *FastAPI*. [Online]. Available: <https://fastapi.tiangolo.com/>. [Accessed: Mar. 5, 2026].

[7] PostgreSQL Global Development Group, "PostgreSQL: Documentation," *PostgreSQL*. [Online]. Available: <https://www.postgresql.org/docs/>. [Accessed: Mar. 5, 2026].

[8] Vercel, "Vercel Documentation," *Vercel*. [Online]. Available: <https://vercel.com/docs>. [Accessed: Mar. 6, 2026].

[9] Supabase, "Supabase Documentation," *Supabase*. [Online]. Available: <https://supabase.com/docs>. [Accessed: Mar. 6, 2026].

Individual Contribution

Team Members	Responsibilities	Key Tasks
MENH Chanheng	Backend Lead	Database Design, System Architecture Design, API development, Security
PIN Sovanda	Frontend Lead	UI/UX design, Client-side logic, Dashboard Integration
HENG Shito	QA, Documentations	Support Frontend, System Testing. Writing the report, User Manual